

LAKAN SINGH DEGAVATH

AI/ML & Software Engineer | LLM Systems | Deep Learning Inference | Distributed Systems | Performance Optimization
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PROFESSIONAL SUMMARY

Software and AI/ML Engineer with 7 years of experience designing, developing, evaluating, and deploying production-grade machine learning and AI systems. Experienced with Python, NLP, LLMs, embeddings, transformers, PyTorch, TensorFlow, model inference, distributed systems, and scalable backend services. Built production AI applications and evaluation workflows with strong experience in model benchmarking, performance analysis, debugging, automated testing, monitoring, and reliability. Experienced in analyzing model behavior and evaluating ML and LLM approaches to build reliable, traceable, and production-ready AI solutions.

TECHNICAL SKILLS

Programming: C++, Python, C, Java, Scala, Go, SQL, JavaScript/TypeScript

Deep Learning & AI: PyTorch, JAX, TensorFlow, Keras, Hugging Face, Machine Learning, Deep Learning, Neural Networks, Deep Neural Networks, LLMs, Generative AI, NLP, scikit-learn, XGBoost

NLP & LLM Systems: Transformers, Embeddings, Semantic Search, RAG, LangChain, LangGraph, Hugging Face, Vector Databases, Prompt Engineering, Context Engineering

AI Evaluation, Safety & Reliability: LLM Evaluation, Model Evaluation, Evaluation Frameworks, Evaluation Pipelines, Model Benchmarking, Model Behavior Analysis, Failure-Mode Analysis, Edge-Case Testing, AI Safety, Trust & Safety, Compliance, Error Analysis, Model Monitoring, Automated Testing, Debugging, Production Reliability

LLM & Model Serving: LLM Inference, Model Serving, Inference Backends, vLLM, SGLang, Quantization, Batching, Low-Latency Inference, Performance Optimization

Computer Architecture & Performance: CPU Architecture, GPU Architecture, x86-64, ARM Architecture, Multicore Processing, Parallel Computing, Cache Hierarchy, Memory Architecture, SIMD/Vectorization, Multithreading, CUDA, GPU Acceleration, GPU Profiling, Memory Optimization, Hardware-Aware Optimization

Inference & Systems: CPU Inference, GPU Inference, Deep Learning Inference, LLM Inference, Model Serving, Inference Backends, vLLM, SGLang, Quantization, Dynamic Batching, Low-Latency Systems, Distributed Systems, Concurrency, Asynchronous Programming, Performance Analysis, Performance Optimization, Fault Tolerance, High Availability.

Software Engineering: Python Development, REST APIs, FastAPI, Microservices, Software Design, System Design, Scalable Backend Services, Distributed Systems, Debugging, Testing, Unit Testing, Integration Testing, Agile/scrum, Code Reviews, Git, GitHub, Linux, CMake, Jira.

AI Infrastructure & DevOps: Docker, Kubernetes, MLflow, Jenkins, CI/CD, Experiment Tracking, Model Monitoring, Logging, Tracing, Observability.

Distributed Computing: PySpark, Apache Spark, Spark Streaming, Kafka, Dask, Flink, Pandas, NumPy

Cloud: AWS, SageMaker, S3, EC2, Lambda

PROFESSIONAL EXPERIENCE

Client: Easybee

Role: AI/ML Engineer

United States · Feb 2026 - Present

- Evaluated and debugged production AI/LLM workflows through automated testing, edge-case testing, failure-mode analysis, performance analysis, monitoring, and systematic validation, identifying unexpected model behavior and reliability issues before production deployment.
- Architected and deployed production-grade Retrieval-Augmented Generation (RAG) applications using LangGraph, LangChain, OpenAI APIs, FastAPI, embeddings, and vector databases, enabling low-latency enterprise knowledge retrieval.
- Designed, developed, and optimized low-latency model inference services and scalable inference backend components for real-time AI applications, applying distributed systems, concurrency, fault tolerance, debugging, and performance optimization techniques.

- Profiled and optimized CPU- and GPU-based model inference workloads, tuning batching, quantization, concurrency, memory utilization, and serving configurations to improve latency, throughput, and resource efficiency.
- Partnered with ML researchers, data scientists, and software engineers to produce AI models through system design, code reviews, automated testing, debugging, performance analysis, and production optimization.
- Designed end-to-end AWS AI architectures using Amazon SageMaker, S3, Lambda, REST APIs, deployment, monitoring, and lifecycle management to support scalable production of AI applications.
- Built reusable MLOps and LLMOps platforms using Docker, CI/CD, model versioning, monitoring, and experiment tracking to streamline AI development and deployment workflows.
- Developed Jenkins CI/CD pipelines to automate unit testing, container builds, model validation, and deployment of AI services across development and production environments.
- Optimized PySpark and streaming data pipelines supporting production machine learning workloads, improving scalability, reliability, and processing efficiency across distributed cloud environments.
- Managed engineering tasks, defects, sprint deliverables, and production issues using Jira within Agile development workflows, coordinating releases across ML, data, and backend teams.

Avila University

Graduate Research Assistant

United States · Apr 2024 - Dec 2025

- Built a reproducible Python NLP and machine learning pipeline for 205K+ product reviews, implementing data preprocessing, text normalization, feature engineering, model training, and evaluation for large-scale sentiment classification.
- Designed and executed model evaluation and benchmarking pipelines for Random Forest, Naive Bayes, and SVM using accuracy, precision, recall, and F1 score, establishing reproducible baselines for evaluating ML and LLM approaches across 205K+ product reviews.
- Evaluated GPT-4 through the OpenAI API for zero-shot and mixed-sentiment classification, analyzing model behavior, edge cases, prediction errors, and performance trade-offs against traditional machine learning baselines.
- Analyzed neural network architectures—including CNNs, RNNs, Bi-LSTMs, and Transformers—to assess their suitability, computational requirements, and performance trade-offs for large-scale sentiment classification.
- Surveyed deep learning and NLP architectures including BERT, CNN/RNN/Bi-LSTM, Llama 2, Word2Vec, GloVe, FastText, and ELMo, and co-authored and presented the resulting research to faculty and peers.

Client: Shopify

Role: Software Engineer

Hyderabad, India · Sep 2020 - Dec 2023

- Designed and built a production real-time machine learning inference platform processing 3M+ financial transactions, integrating model inference, distributed backend components, Kafka, and Spark Streaming into scalable production pipelines.
- Designed, analyzed, and optimized distributed streaming and model-inference pipelines, identifying and resolving performance bottlenecks to reduce near-real-time detection latency by 45%.
- Profiled CPU utilization, memory consumption, thread concurrency, and data-movement bottlenecks across distributed inference workloads, applying multithreading and memory optimization techniques to improve processing efficiency.
- Optimized machine learning models using ROC-AUC, Precision-Recall analysis, SHAP, and LIME, reducing false positives to fewer than 2 per 1,000 transactions while improving model explainability.
- Built reusable inference backend and distributed-processing components using PySpark and Dask, improving scalability, reliability, and operational efficiency at production scale.
- Evaluated CPU- and GPU-based inference configurations, comparing latency, throughput, memory utilization, and deployment-cost trade-offs for production model-serving workloads.
- Conducted A/B testing comparing rule-based and machine learning-based fraud detection strategies, deploying the higher-performing solution and contributing to the prevention of over \$90K in annual fraud losses.
- Collaborated with data scientists, ML engineers, backend engineers, and product teams to produce fraud detection solutions through system design, code reviews, automated testing, debugging, and production performance optimization.

Associate Software Engineering

Hyderabad, India · Sep 2019 - Aug 2020

- Developed and maintained C++, Java, and Scala components within a large enterprise codebase, applying object-oriented design, code reviews, automated testing, debugging, and production engineering practices to build reliable and maintainable system software.
- Configured and maintained C/C++ build workflows using CMake, managing dependencies, compiler settings, build targets, and automated unit-test integration.

- Analyzed C/C++ application performance across x86-64 and multicore CPU environments, applying profiling, cache-aware programming, multithreading, and memory-management techniques to improve runtime efficiency.
- Implemented and tested hardware-aware optimizations using SIMD/vectorization and multicore parallelism across x86-64 and ARM-based environments.

PROJECTS

GPU-Accelerated LLM Inference & Performance Optimization

- Designed and implemented GPU-accelerated LLM inference pipelines using PyTorch, CUDA, TensorRT-LLM, and vLLM on NVIDIA GPUs, benchmarking latency, throughput, and GPU memory utilization across inference backends.
- Benchmarked CPU and GPU inference across representative model sizes and batch configurations, comparing latency, throughput, memory utilization, and hardware efficiency to identify suitable serving configurations.
- Optimized TensorRT-LLM inference configurations through batching, quantization, and engine-build parameter tuning, analyzing latency, throughput, and GPU-memory trade-offs for efficient LLM inference.
- Profiled GPU workloads using NVIDIA profiling tools, investigating kernel execution, memory transfers, compute utilization, and synchronization overhead to identify performance bottlenecks.
- Analyzed GPU architecture characteristics—including streaming multiprocessors, memory hierarchy, tensor cores, thread blocks, and warp execution—to guide inference configuration and hardware-aware optimization decisions.
- Evaluated CPU architecture characteristics, including cache hierarchy, vectorization, thread scheduling, and memory bandwidth, when optimizing preprocessing and CPU inference workloads.
- Analyzed CUDA execution, GPU memory management, parallel computation, and kernel-level performance characteristics through close-to-metal performance analysis for hardware-aware deep learning inference optimization.
- Implemented GPU compiler optimization workflows using MLIR and LLVM, applying computational graph transformations and compiler passes to optimize execution paths for GPU-accelerated deep learning inference.

Enterprise Multi-Agent AI Platform

- Designed scalable LLM inference and retrieval services using Python, FastAPI, LangGraph, FAISS, and distributed backend components, supporting concurrent requests and low-latency enterprise knowledge retrieval.
- Implemented modular retrieval and inference workflows supporting document ingestion, embeddings, context assembly, tool execution, routing, memory, and LLM response generation.
- Built containerized model-serving APIs with Docker, monitoring, logging, automated testing, and deployment on AWS, emphasizing reliability and production performance.
- Implemented Jenkins-based CI/CD workflows for unit testing, integration testing, Docker image creation, and automated deployment of model-serving services.

EDUCATION

Avila University

M.S., Computer Science – AI/ML Specialization

United States

• **GPA:** 3.91/4.0

• **Coursework:** Distributed Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Data Structures & Algorithms, Computer Networks, Software Engineering, Big Data Analytics